

Welcome to 42417 Simulation in Operations Management!

The course will take place during the period between June 8th and June 26th. The course will be taught at the DTU campus in Kgs. Lyngby in building 358, room 060a+060b. All information concerning the course is communicated via DTU Learn.

Course plan – TENTATIVE! Changes will be communicated via DTU Learn!

Date	Topic	Literature + media
Mon 08-June	09:00 Lecture: Intro. to the course and formalities 10:00 Lecture: Intro. to simulation & simulation modeling 13:30 Lecture: Introduction simulation tools and AnyLogic 14:30 Exercises using AnyLogic 16:45 Wrap-up of the day	[Slides + course plan] [Ros. 1.1->1.7 + 4.4->4.5.2] [AnyLogic p.7->14] [AnyLogic Bank office tutorial] -
Tues 09-June	09:00 Lecture: Simulation modeling (continued) 09:30 Lecture: Bottleneck ana. + performance management 11:00 Exercises using AnyLogic 15:00 Introduction to AnyLogic Airport case 16:45 Wrap-up of the day	[Slides] [Ros. 1.1->1.7 + 4.4->4.5.2] [Exercise description + CourseNotes + eLearning] [Exercise description] -
Wed 10-June	09:00 Lecture: Modeling, implementation and verification, calibration and validation 10:00 Guest lecture: "Simulation modelling in real-world applications" by NIRAS 11:00 AnyLogic Airport case 16:45 Wrap-up of the day	[Ros. 1.1->1.7] [CourseNotes + eLearning] - [Exercise description] -
Thurs 11-June	09:00 Lecture: Input data modeling 10:30 Exercises in input modeling 13:00 AnyLogic Airport case 16:45 Wrap-up of the day	[Ros. 6.1->6.4] [CourseNotes + eLearning] [Exercise description] -
Fri 12-June	09:00 Lecture: Output analysis + experimental design 10:30 AnyLogic Airport case 16:00 Prize competition + Wrapping-up before exam	[Ros. 7.1->7.2 + 2.1->2.3] + [CourseNotes 5.1->5.2.4] + [CourseNotes + eLearning] All of the above mentioned
Mon 15-June	10:00-11:00 Written midterm exam – mandatory physical presence! 13:00 Lecture: Presentation of cases and requirements to projects	
Tues 16-June	09:00 Q&A session – first questions to the case study projects After: Group work on case study project	
16-June until 26-June	All day: Group work on case study project 18:00 Deadline for hand-in of written report	

Literature + media (uploaded to DTU Learn)

[AnyLogic] – Ilya Grigoryev, “AnyLogic in Three Days, A quick course in simulation modeling”, 6th edition, 2024.

[Ros X.X] – Rossetti, Manuel: “Simulation Modelling and Arena”, Chapter X.X, Wiley, 2016. Electronic version accessible at: <https://findit.dtu.dk/en/catalog/2484023712>

[CourseNotes X.X] – “42417 Simulation in Operations Management – Course Notes”, DTU Management, 2026.

Besides the above, additional literature etc. will be uploaded to DTU Learn.

Lecturers and Teaching Assistants (TAs)

Allan Larsen, Professor, DTU Management, e-mail: alar@dtu.dk

Anneke Claypool, Assistant Professor, DTU Management, alaci@dtu.dk

Cornelius Valentin Hasselriis, PhD student, DTU Management, cvaha@dtu.dk

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Catrina Pietralla, PhD student, e-mail: catpie@dtu.dk

The TAs will be available for questioning based on a schedule. The TAs help you learn simulation modeling, but please do not expect the TAs to solve all your problems. We strongly encourage you to share experiences with your co-students to make more efficient use of the TA resources.

Course structure and evaluation

The course demands active participation from the students. It is expected that the students will read and reflect upon the course literature and make an effort to become familiar with the software used in the course.

During the three-week period, teaching activities will consist of introductory lectures, small exercises and a larger case study project.

Monday, June 15th, all students must participate in a **mandatory written midterm exam**. The midterm exam evaluates each student's understanding of the basic simulation theory covered in the first week of teaching.

Following the midterm exam on **June 15th**, the case study projects will be introduced to all students. The case study project is required to be carried out in **groups of two students**. The requirement to work in groups of two students is rigid and is based on previous experiences with teaching modelling, implementation and analysis using software and will increase learning outcomes. The specific details of the case study project offerings will be provided in a separate document to be distributed on June 15th. The case study project must be handed in electronically via DTU Learn no later than **Friday, June 26th at 18:00**.

The final course grade will be based on an overall assessment of the student's ability to cover the learning objectives of the course and will therefore be based on the assessment from the written midterm exam and the assessment of the case study project. Also, it is required that the student passes the written midterm exam and the case study.